

CMPG 325 - Computer Networks

Milestone 2

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CLIENT IMPLEMENTATION REVIEW

A. EXECUTIVE SUMMARY & SYSTEM OVERVIEW

I. PURPOSE AND SCOPE

The network has been designed, implemented, and fully verified within Cisco Packet Tracer based on the assigned network base 192.168.55.0/24**.** The architecture addresses all operational requirements for an entertainment venue:

- **High-Speed Gaming Lounge:** Support for interactive gaming stations on VLAN 10.
- **Administrative Operations:** Workstations for lounge management on VLAN 20.
- **CCTV Security Subsystem**: Webcams placed on VLAN 30 and isolated from office data traffic.
- **Network Printing:** Shared printer on VLAN 70 accessible to staff.
- **Visitor Infrastructure (CR3):** Wireless guest Wi-Fi on VLAN 40 isolated from internal resources.
- **Core Network Services:** Centralized DHCP/DNS server and external Internet server hosting the Leano web portal.

II. ASSIGNED TECHNICAL CHALLENGE & CHANGE REQUEST REVIEW

- **Technical Challenge: VLAN Trunking (802.1Q Across Switches)**
    - **Objective:** Allow endpoints physically attached across two switches (SW1 and SW2) to belong to the same logical subnets while maintaining isolation.
    - **Implementation Summary:** The link between SW1 (Fa0/24) and SW2 (Fa0/24) is configured as an 802.1Q trunk. Packets sent from CAM3-CAM5 or Gaming PC3-PC5 on SW2 are tagged with their respective VLAN IDs before crossing to SW1 and routed via R1 subinterfaces.

- **Design Constraint: CCTV Traffic Segmentation**
    - **Objective:** CCTV traffic must be strictly segmented from office data traffic.
    - **Implementation Summary:** Cameras are assigned to VLAN 30 (192.168.55.32/28). The extended access list CCTV-IN applied to subinterface Gig0/0.30 permits cameras to reach only their gateway and the Internal Server subnet, explicitly blocking traffic to office workstations.

- **Change Request 3 (CR3): Guest Wi-Fi Access**
    - **Objective:** Provide Wi-Fi connectivity for visitors while isolating them completely from internal lounge assets.
    - **Implementation Summary:** Access Point Leano-Guest is connected to SW2 (Fa0/10) on VLAN 40 (192.168.55.48/28). Router R1 provides local DHCP leases. Extended access list GUEST-IN applied on subinterface Gig0/0.40

permits outbound Internet and local DHCP while dropping all packets destined for internal VLANs.

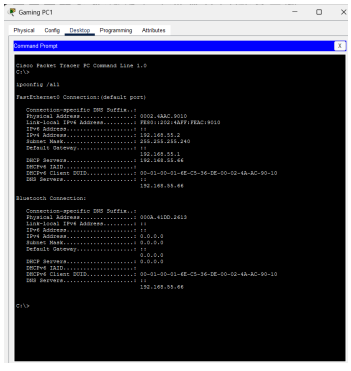
B. ACCESS CONTROL LIST (ACL) SECURITY POLICIES

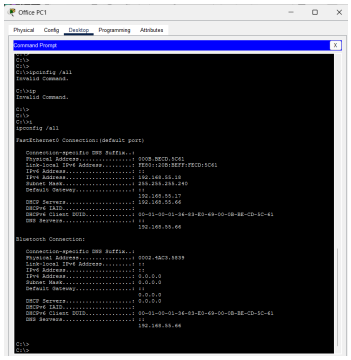
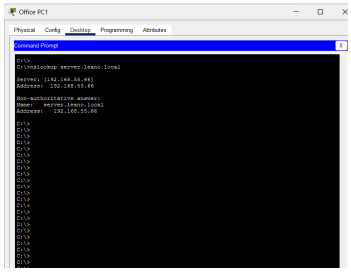
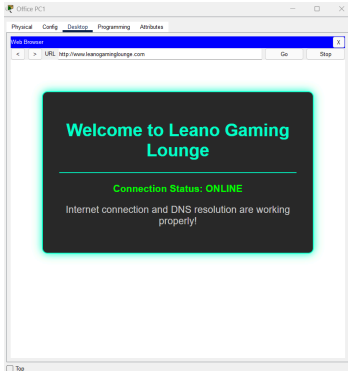
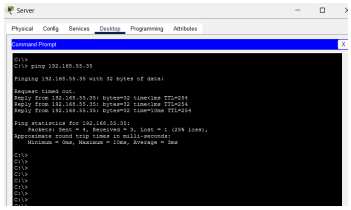
I. ROUTER R1 APPLIED SECURITY POLICY MATRIX

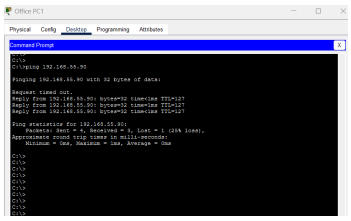
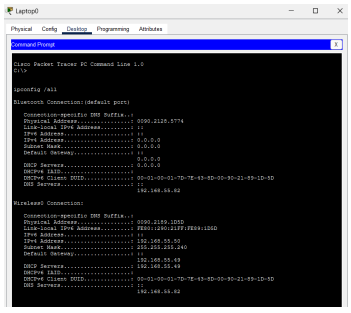
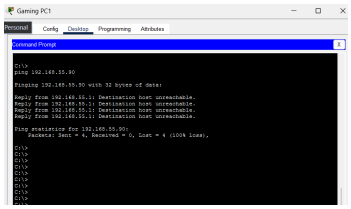
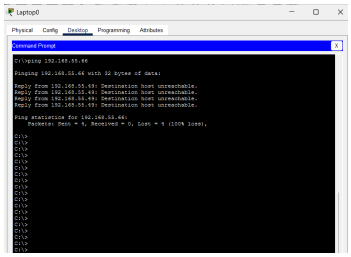
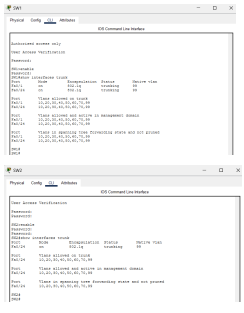
ACL Name	Interface / Subinterface	Target Direction	Applied Rule & Function
CCTV-IN	GigabitEthernet0/0.30	Inbound	Permit: IP traffic to Default Gateway (192.168.55.33) and Server Subnet (192.168.55.64/29) Deny: All IP traffic to Office network (192.168.55.16/28)
PRINTER-IN	GigabitEthernet0/0.70	Inbound	Permit: Traffic from Printer to Office subnet (192.168.55.16/28) and Gateway (192.168.55.89) Deny: All other traffic (blocks Gaming & Guest subnets)
OFFICE-IN	GigabitEthernet0/0.20	Inbound	Deny: IP traffic directed to CCTV network (192.168.55.32/28) • Permit: All other traffic (allows access to Printer, Internal Server, Internet)
GAMING-IN	GigabitEthernet0/0.10	Inbound	Deny: IP traffic directed to CCTV (192.168.55.32/28) and Printer (192.168.55.88/29)

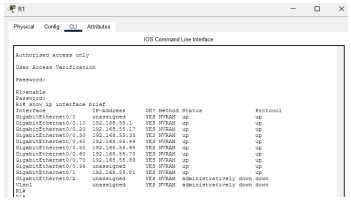
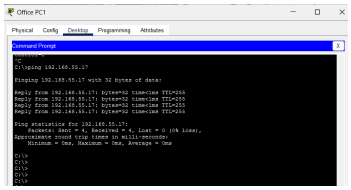
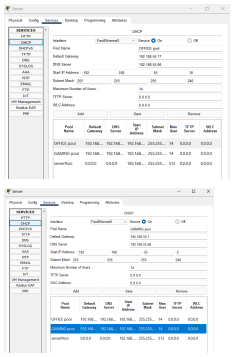
			• Permit: All other traffic (allows access to Internal Server & Internet)
GUEST-IN	GigabitEthernet0/0.40	Inbound	Permit: UDP DHCP traffic (ports 67/68) and outbound Internet access Deny: IP traffic to Gaming, Office, CCTV, Server, and Printer subnets

C. TESTING EVIDENCE & VERIFICATION MATRIX

Test ID	Category	Action / Test Description	Expected Result	Status	Evidence Reference
T01	DHCP Lease	Configure Gaming PC1 to DHCP.	Obtains IP in range 192.168.55.2–14 with DNS 192.168.55.66	PASS	

T02	DHCP Lease	Configure Office PC1 to DHCP.	Obtains IP in range 192.168.55.18–30 with gateway 192.168.55.17	PASS	
T03	Internal DNS	Execute nslookup server.leano.local on Office PC1.	Resolves successfully to 192.168.55.66	PASS	
T04	HTTP Web	Open www.leanogaminglounge.com from the Office PC browser.	Leano Gaming Lounge landing page renders successfully	PASS	
T05	Static IP	Ping CAM4 (192.168.55.35 on SW2) from Internal Server	Ping replies successfully across trunk link	PASS	

T06	Printer Reach	Ping Network Printer (192.168.55.90) from Office PC1.	Successful ping replies confirming cross-VLAN printing	PASS	
T06	Guest Wi-Fi	Connect Wireless Laptop to SSID Leano-Guest	Receives IP 192.168.55.50 from gateway 192.168.55.49	PASS	
T08	CCTV ACL	Ping Office PC1 (192.168.55.18) from CAM1 (192.168.55.34).	Blocked by CCTV-IN ACL (Destination Host Unreachable)	PASS	
T09	Gaming ACL	Ping Network Printer (192.168.55.90) from Gaming PC1.	Blocked by GAMING-IN ACL	PASS	
T10	Guest ACL	Ping Internal Server (192.168.55.66) from Guest Laptop.	Blocked by GUEST-IN ACL enforcing complete isolation	PASS	
T11	Trunking	Run show interfaces trunk on SW1 and SW2 CLI.	Inter-switch link (Fa0/24) status shows trunking under 802.1Q	PASS	

T12	ROAS Status	Run show ip interface brief on R1 CLI.	Subinterfaces Gig0/0.10 to .99 display up/up	PASS	
T13	Inter-VLAN	Ping Office PC3 (192.168.55.17) from Office PC1 (192.168.55.19).	Successful ICMP communication across switches	PASS	
T14	DHCP Binding	View active lease table on Internal Server GUI.	Logs active MAC-to-IP leases for Gaming and Office PCs	PASS	
T15	NVRAM Backup	Run copy running-config startup-config on R1, SW1, SW2.	Returns [OK]; configuration written to startup memory	PASS	

D. SOFTWARE TOOL AND PROJECT PROFILE

I. **Cisco Packet Tracer:** The primary simulation environment used to build the physical and logical topology, configure the 802.1Q trunking protocols, and perform end-to-end connectivity testing.

II. **Repository Link:** https://github.com/MateyeSelota/CMPG325-2026-125